2 **(a)** State Newton's second law of motion.

The rate of change of momentum of an object is directly proportional to the resultant force acting on it, and takes place in the direction of the force. [1]

(b) A ball of mass 150 g travelling horizontally at a speed of 20 m s^{-1} collides with a vertical wall. Fig. 2.1 shows how the force of impact on the wall, F, varies with time t.

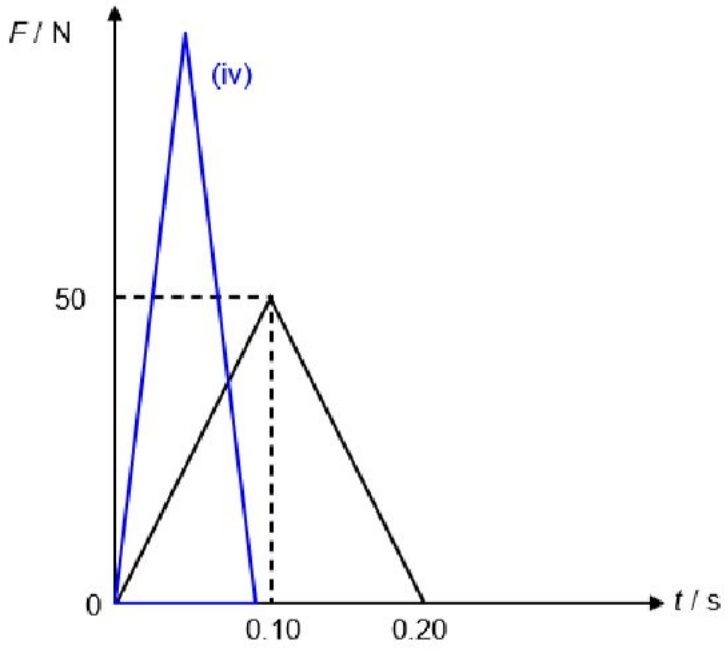


Fig. 2.1

(i) Determine the change in momentum of the ball.

Change in momentum = area under graph [1]

$$= \frac{50 \times 0.20}{2}$$

$$= 5.0 \text{ N s} \quad [1]$$

change in momentum = ... N s [2]

(ii) Hence determine the final speed of the ball.

Take rightwards as positive.
(Note: the impact force on the ball is to the left. So change in momentum of the ball is also to the left i.e. negative.)

Change in momentum = $m(v_f - v_i)$

$$-5.0 = 0.150(v_f - 20)$$

$$v_f = -13.33 \text{ m s}^{-1}$$

$\therefore$ Speed = 13 m s^{-1}

final speed = ... m s^{-1} [1]

(iii) Balls which are identical to the one above are thrown at the wall at the rate of 30 balls per minute. Determine the average impact force on the wall.

$$\text{Average force} = \frac{\text{Total change in momentum of 30 balls}}{\text{time interval}} \quad [1]$$

$$= \frac{30 \times 5.0}{60}$$

$$= 2.5 \text{ N} \quad [2]$$

average impact force = ... N [2]

(iv) A ball which is equal in mass as the one above is made of a harder material. It also collides with the wall at the same speed as before. Draw on Fig. 2.1 a second graph to show how the impact force on the wall varies with time when this ball collides with the wall. [1]

Higher peak and shorter time interval [1]

[Total: 7]